North Bengal International University

Lab Report: 08

Exp Name : implementation the enqueue and dequeue operation using C language.

Subject : Data Structure

Subject Code : CSE 0612-2131

Submitted By:

Name : Md. Tanvir Rahman Pranto
Dept : CSE
Roll : 0712340012101005
Semester : 2nd Year, 4th semester.

Submitted To:

Name : Saifur Rahman
Designation : Lecturer
Dept : CSE
Date :

Code:

```c
#include <stdio.h>
#define SIZE 5

int main() {
    int queue[SIZE];
    int front = -1;
    int rear = -1;
    int choice, value, i;

    while (1) {
        printf("\n1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            if (rear == SIZE - 1) {
                printf("Queue is full!\n");
            } else {
                printf("Enter value to enqueue: ");
                scanf("%d", &value);
                if (front == -1 && rear == -1) {
                    front = 0;
                    rear = 0;
                } else {
                    rear = rear + 1;
                }
                queue[rear] = value;
                printf("Enqueued: %d\n", value);
            }
        }

        else if (choice == 2) {
            if (front == -1 || front > rear) {
                printf("Queue is empty!\n");
            } else {
                printf("Dequeued: %d\n", queue[front]);
                front = front + 1;
            }
        }

        else if (choice == 3) {
            if (front == -1 || front > rear) {
                printf("Queue is empty!\n");
            } else {
                printf("Queue elements: ");
                for (i = front; i <= rear; i++) {
                    printf("%d ", queue[i]);
                }
                printf("\n");
            }
        }

        else if (choice == 4) {
            break;
        }
```

```
        else {
            printf("Invalid choice\n");
        }
    }

    return 0;
}
```

Output:

```
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value to enqueue: 10
Enqueued: 10

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value to enqueue: 20
Enqueued: 20

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value to enqueue: 30
Enqueued: 30

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 10 20 30
```

Conclusion: This lab helped me understand how a queue works using simple C code. I learned how to add (enqueue) and remove (dequeue) elements step by step using only basic logic and variables. This practice improved my understanding of data structures without using advanced techniques.